# High-Speed Link Design

**Lab Report 03**

---

# Low & High-Swing Voltage-Mode Driver Design

---


**Made By:**

Hadey Sayed Mohamed

**Department of Electronics & Electrical Communication Engineering**


Cadence Virtuoso Simulation Analysis

# Contents

# List of Figures



# List of Tables

# 1 Low-Swing Voltage Mode Driver

## 1.1 Objective

The objective of this part is to design and characterize a low-swing voltage-mode driver suitable for high-speed serial-link applications. The driver operates from a supply voltage of

$$V_{DD} = 500 \text{ mV}, \tag{1}$$

and is operated at a data rate of 8 Gb/s. The driver is implemented using an NMOS-under-NMOS (NUN) differential structure. The transistor sizing is optimized using an S-parameter testbench, after which the optimum sizes are used in the complete driver to evaluate its transient response, differential output swing, eye diagram, and average power consumption.

The output loading includes a 1 pF capacitor at each output node, representing the ESD and pin capacitance specified in the lab.

## 1.2 Low-Swing Driver Architecture

The low-swing driver consists of four NMOS transistors arranged as two output branches. The pull-up devices are $M_1$ and $M_0$, while the pull-down devices are $M_2$ and $M_3$.

The two output branches can be represented as

$$V_{DD} \to M_1 \to V_{OUTP} \to M_2 \to GND \tag{2}$$

and

$$V_{DD} \to M_0 \to V_{OUTN} \to M_3 \to GND. \tag{3}$$

The gates are cross-controlled by complementary input signals:

$$M_1 : VINP, \qquad M_2 : VINN, \tag{4}$$

$$M_0 : VINN, \qquad M_3 : VINP. \tag{5}$$

The differential output is defined as

$$V_{ODIFF} = V_{OUTP} - V_{OUTN}. \tag{6}$$

When $VINP = 1$ and $VINN = 0$, $M_1$ and $M_3$ are ON while $M_0$ and $M_2$ are OFF. Consequently, $V_{OUTP}$ is driven toward the high level while $V_{OUTN}$ is driven toward the low level, producing a positive differential output. When the complementary input state occurs, the roles of the two branches are reversed and a negative differential output is generated.

The input sources use complementary PRBS signals with

$$T_{\text{UI}} = \frac{1}{8 \text{ Gb/s}} = 125 \text{ ps}, \tag{7}$$

with zero and one levels of 0 V and 1 V, respectively. The rise and fall times are 6.25 ps.

Table 1: PRBS input-source settings for the low-swing driver.

| Parameter | Value |
|---|---|
| Zero value | 0 V |
| One value | 1 V |
| Bit period | 125 ps |
| Rise time | 6.25 ps |
| Fall time | 6.25 ps |
| Edge type | Linear |
| LFSR mode | PN32 |
| Input relationship | $VINN = \overline{VINP}$ |


Figure 1: Schematic of the low-swing NMOS-under-NMOS voltage-mode driver.

## 1.3 Sizing Testbench

A separate S-parameter testbench was constructed to determine the optimum sizing of the pull-up and pull-down devices. Each transistor was tested individually while being biased in its ON state. The output node was loaded with a 50 Ω port and a 1 pF capacitor to reproduce the high-speed loading conditions of the actual driver.

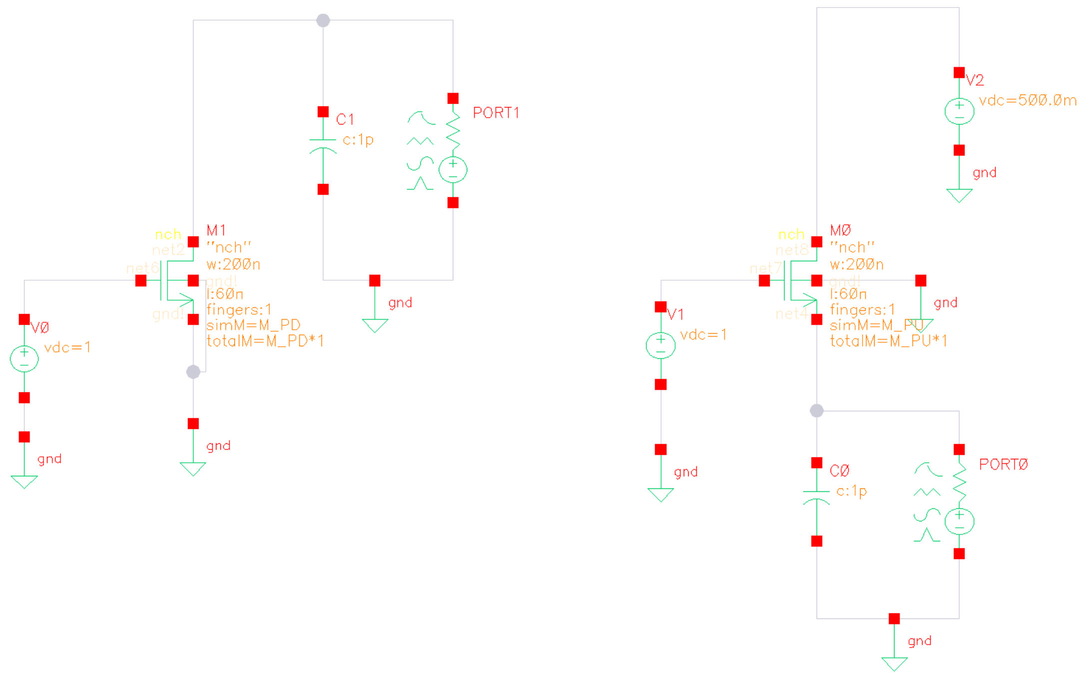


Figure 2: S-parameter sizing testbench for the pull-up and pull-down NMOS devices.

The effective transistor width is controlled through the multiplier $M$. Increasing $M$ increases the effective transistor width and therefore increases the transistor drive strength. However, larger devices also introduce larger parasitic capacitances. Consequently, an optimum device size exists which provides the best high-frequency matching.

The S-parameter analysis was performed at the frequency corresponding to the Nyquist frequency of the 8-Gb/s NRZ data:

$$f_N = \frac{R_b}{2} = \frac{8\ \text{Gb/s}}{2} = 4\ \text{GHz}. \tag{8}$$

Therefore, the quantity used to compare the different transistor sizes was

$$S_{11,\text{dB}}(4\ \text{GHz}). \tag{9}$$

The optimum multiplier is the value of $M$ that gives the minimum (most negative) value of $S_{11}$ at 4 GHz.

## 1.4 Pull-Down Device Characterization

The pull-down sizing testbench was used to characterize the NMOS pull-down device. The multiplier was swept from

$$M = 10 \rightarrow 150 \tag{10}$$

with a step of one, as specified by the lab procedure.

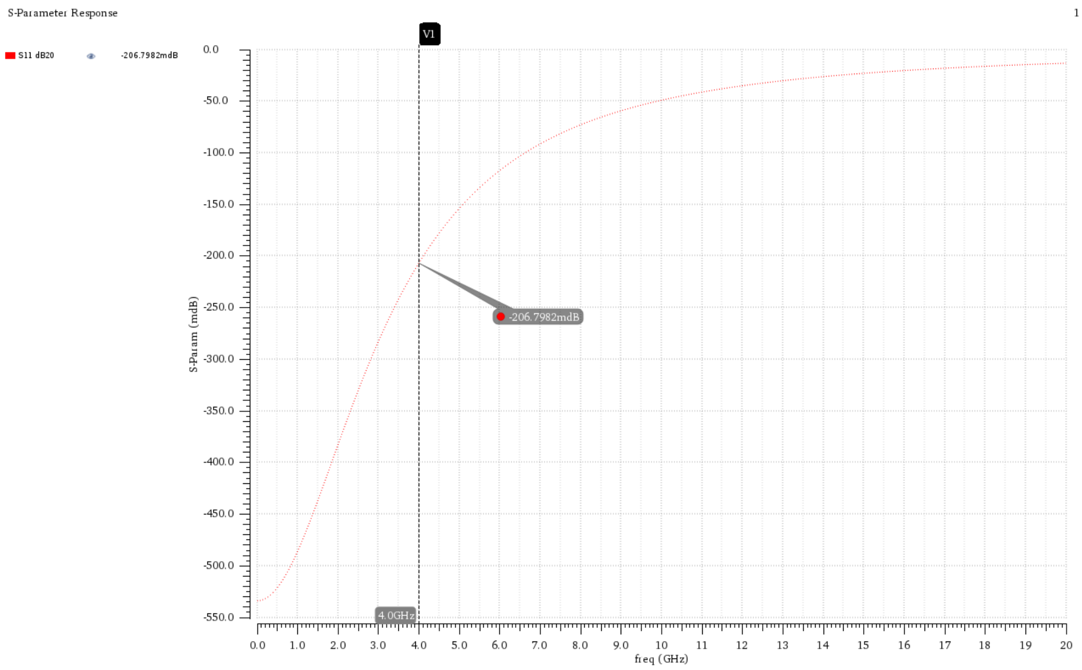


Figure 3: $S_{11}$ versus frequency for the pull-down sizing testbench.

The simulated $S_{11}$ response generally becomes less negative as the frequency increases, indicating degradation of the impedance matching at higher frequencies. This behavior is mainly attributed to the increasing effect of the transistor and load parasitic capacitances, which make the input impedance increasingly frequency-dependent.

The multiplier sweep is shown in Figure 4.

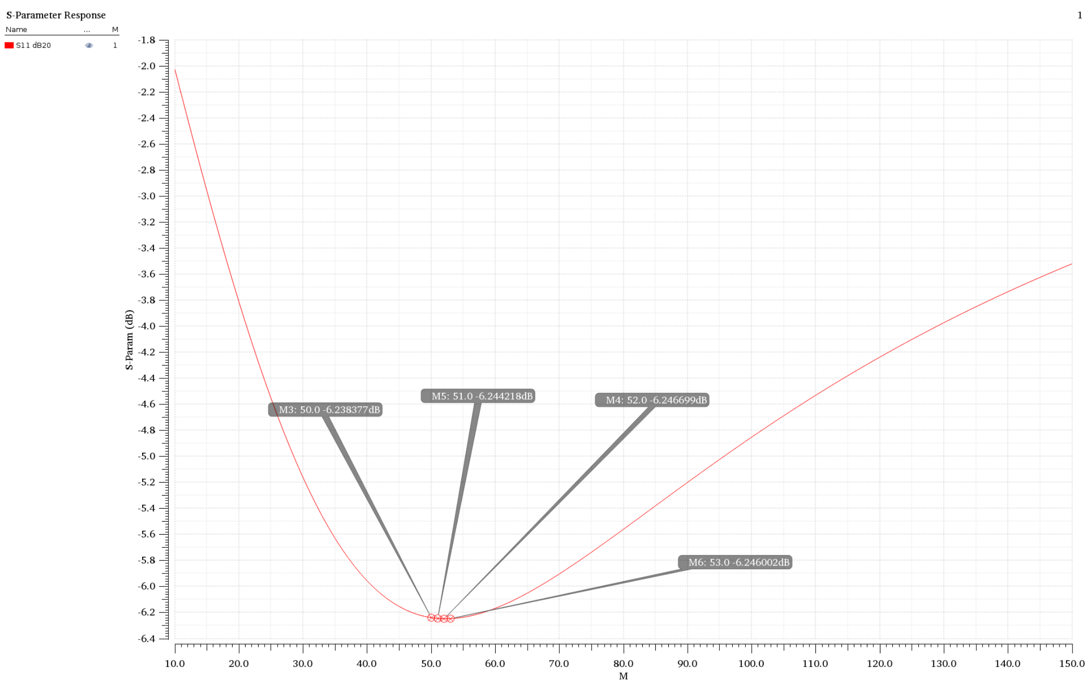


Figure 4: $S_{11}$(4 GHz) versus multiplier $M$ for the pull-down device.

The $S_{11}$ versus $M$ curve exhibits a clear minimum. For small values of $M$, the transistor is relatively weak, resulting in a relatively poor high-frequency match. Increasing $M$

improves the matching until an optimum is reached. Further increasing the transistor size causes the parasitic capacitances to become more significant, so the matching starts to degrade again.

The most negative value of $S_{11}$ was obtained at

$$\boxed{M_{PD} = 52} \tag{11}$$

with

$$S_{11}(4\ \text{GHz}) \approx -6.2467\ \text{dB}. \tag{12}$$

The curve is relatively flat around the optimum. For example,

$$M = 51 : -6.2442\ \text{dB}, \tag{13}$$
$$M = 52 : -6.2467\ \text{dB}, \tag{14}$$
$$M = 53 : -6.2460\ \text{dB}. \tag{15}$$

Therefore, $M = 52$ was selected as the optimum pull-down multiplier.

## 1.5 Pull-Up Device Characterization

The same sizing procedure was applied to the pull-up NMOS devices. The multiplier was swept and the $S_{11}$ value was evaluated at 4 GHz.

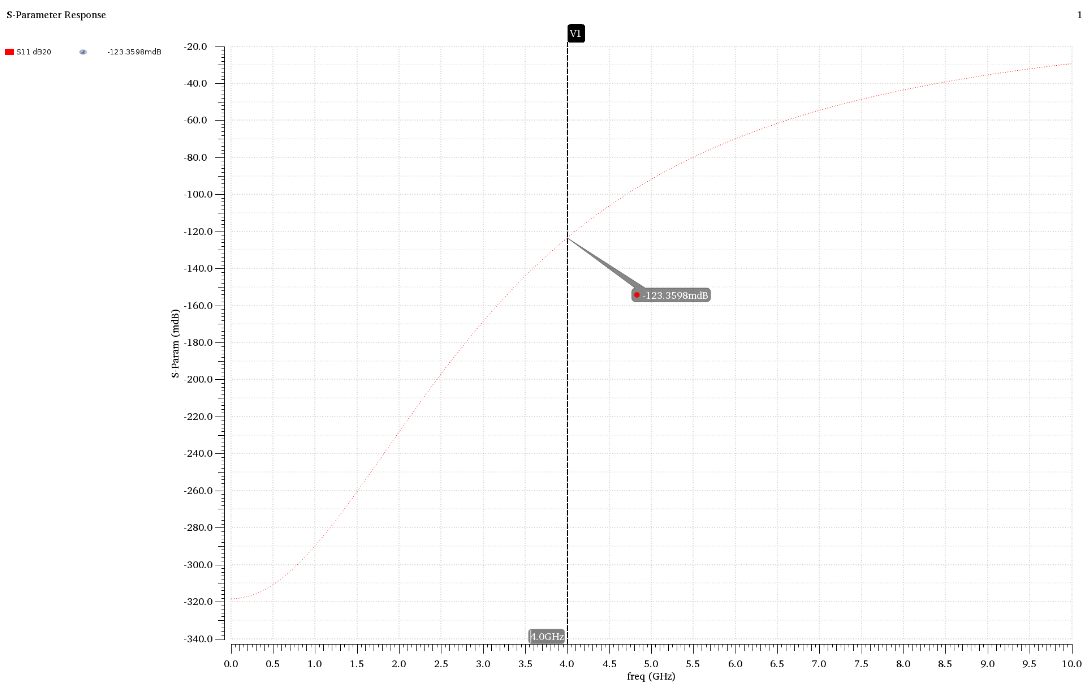


Figure 5: $S_{11}$ versus frequency for the pull-up sizing testbench.

The frequency response shows the same general trend observed for the pull-down device: matching degrades at higher frequencies due to the increasing effect of parasitic capacitances.

The $S_{11}$ versus multiplier result is shown in Figure 6.

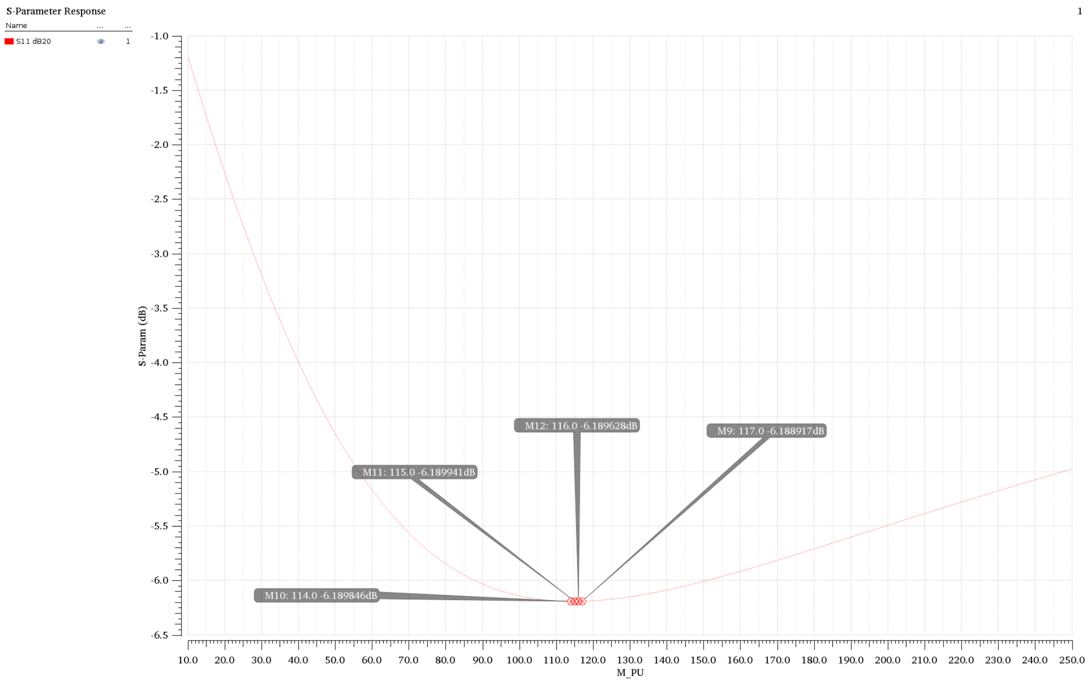


Figure 6: $S_{11}$(4 GHz) versus multiplier $M$ for the pull-up device.

The curve exhibits a minimum similar to the pull-down sizing result. At small multiplier values, the pull-up device is relatively weak and the matching is poor. Increasing the multiplier improves the matching, while further increasing the device size eventually increases the parasitic capacitance enough to degrade the high-frequency response.

The optimum pull-up multiplier obtained from the simulation is

$$\boxed{M_{PU} = 115} \tag{16}$$

with

$$S_{11}(4 \text{ GHz}) \approx -6.1899 \text{ dB}. \tag{17}$$

The minimum is relatively broad around the optimum, but $M = 115$ gives the most negative measured $S_{11}$ and was therefore selected.

## 1.6 Comparison of Pull-Up and Pull-Down Sizing

The optimum transistor sizes obtained from the S-parameter simulations are

$$\boxed{M_1 = M_0 = 115} \tag{18}$$

for the pull-up devices and

$$\boxed{M_2 = M_3 = 52} \tag{19}$$

for the pull-down devices.

The pull-up devices require a larger multiplier because an NMOS used in the pull-up path experiences a reduction in $V_{GS}$ as the output voltage rises. Consequently, its available drive strength decreases near the high output level. A larger device is therefore required to obtain comparable high-frequency behavior.

The difference in optimum sizing also demonstrates that the pull-up and pull-down devices do not experience identical operating conditions and therefore do not necessarily require identical dimensions.

## 1.7 Transient Simulation of the Optimized Driver

After determining the optimum transistor sizing, the values

$$M_1 = M_0 = 115, \qquad M_2 = M_3 = 52 \tag{20}$$

were used in the complete low-swing voltage-mode driver.
A transient simulation was then performed for

$$T_{\text{sim}} = 50 \text{ ns}, \tag{21}$$

as required by the lab procedure.
The differential output was measured as

$$V_{ODIFF} = V_{OUTP} - V_{OUTN}. \tag{22}$$

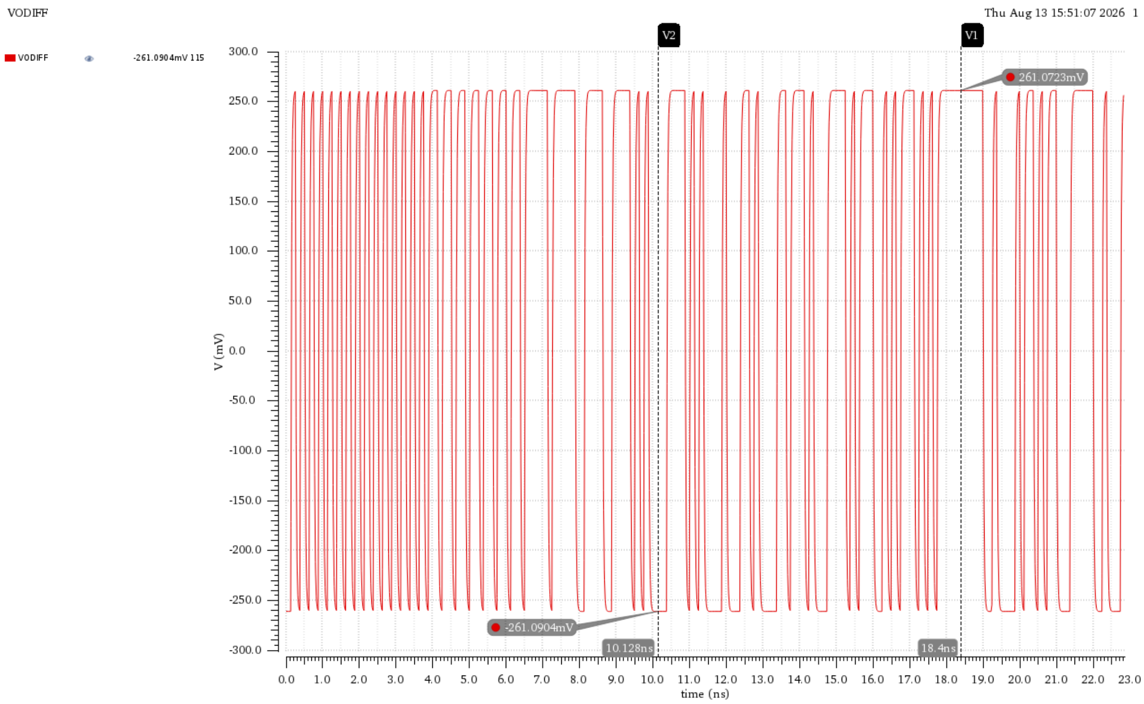


Figure 7: Transient response of the differential output of the optimized low-swing driver.

The simulated differential output reaches approximately

$$V_{ODIFF,\text{max}} \approx +261.07 \text{ mV}, \tag{23}$$
$$V_{ODIFF,\text{min}} \approx -261.09 \text{ mV}. \tag{24}$$

Therefore, the differential peak-to-peak amplitude is

$$V_{ODIFF,pp} = V_{ODIFF,\max} - V_{ODIFF,\min}. \tag{25}$$

Thus,

$$V_{ODIFF,pp} = 261.07 - (-261.09) \approx \boxed{522.16\ \text{mV}_{pp}}. \tag{26}$$

The differential waveform is nearly symmetric around zero. The average of the positive and negative levels is

$$\frac{261.07 + (-261.09)}{2} \approx -0.01\ \text{mV}. \tag{27}$$

This confirms the nearly symmetric differential operation of the two branches.

## 1.8 Eye Diagram

The eye diagram was generated from the differential output using the optimized transistor sizes. The data rate is 8 Gb/s, corresponding to

$$T_{\text{UI}} = 125\ \text{ps}. \tag{28}$$

The eye diagram therefore uses a 125 ps unit interval and overlays successive waveform segments to visualize the signal integrity at the driver output.

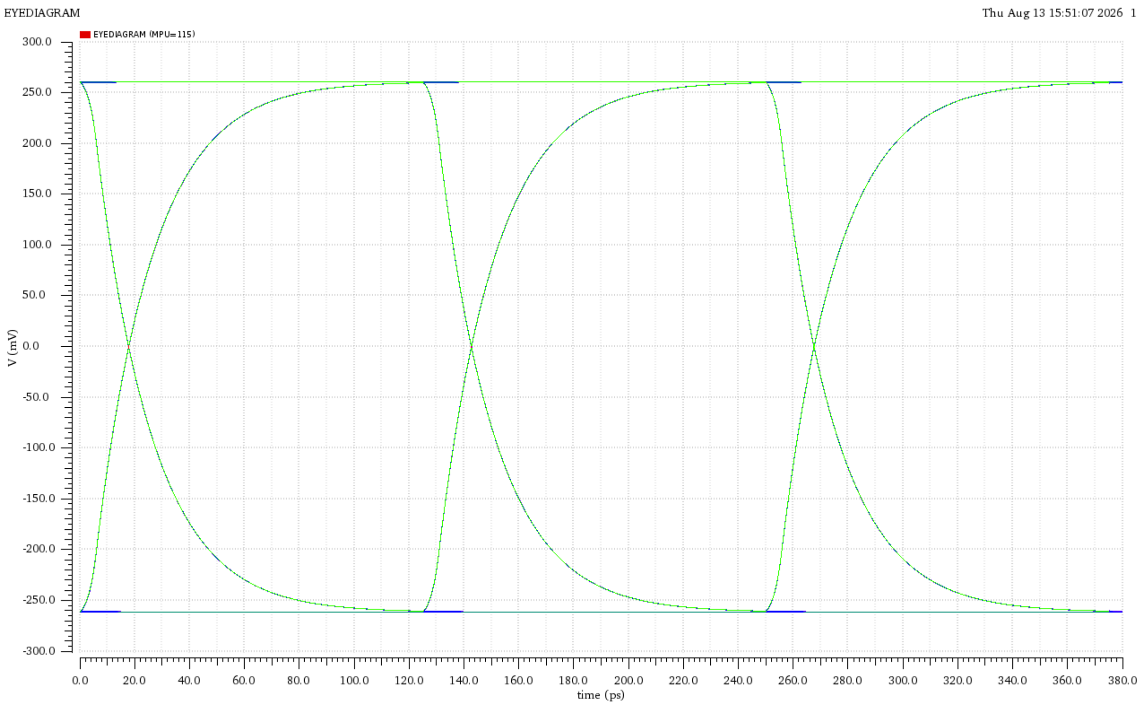


Figure 8: Eye diagram of the differential output of the optimized low-swing voltage-mode driver.

The upper and lower differential levels are approximately

$$V_1 \approx +261\ \text{mV}, \tag{29}$$
$$V_0 \approx -261\ \text{mV}. \tag{30}$$

Hence, the observed vertical eye opening is approximately

$$H_{\text{eye}} = V_1 - V_0 \approx 261 - (-261) \approx \boxed{522 \text{ mV}}. \tag{31}$$

The eye diagram shows a clear and approximately symmetric differential signal. The large vertical opening indicates a substantial voltage margin at the driver output. The symmetry of the upper and lower levels around zero also confirms the differential nature of the output.

## 1.9 Average Power Consumption

The average power was calculated according to the method specified in the lab:

$$P_{\text{avg}} = V_{DD} I_{DD,\text{avg}}. \tag{32}$$

The average current drawn from the $V_{DD}$ supply was obtained from the transient simulation as

$$I_{DD,\text{avg}} = 5.884 \text{ mA}. \tag{33}$$

| | | |
|---|---|---|
| LAB_03:LOW_SWING_VM_DRIVER:1 | ID_avg | 5.884m |
| LAB_03:LOW_SWING_VM_DRIVER:1 | Pavg | 2.942m |
| LAB_03:LOW_SWING_VM_DRIVER:1 | /M1/D | |
| LAB_03:LOW_SWING_VM_DRIVER:1 | /M0/D | |

Figure 9: Supply current waveform and average-current measurement used for the power calculation.

Since

$$V_{DD} = 500 \text{ mV} = 0.5 \text{ V}, \tag{34}$$

the average power is

$$P_{\text{avg}} = (0.5)(5.884 \text{ mA}) \tag{35}$$
$$= \boxed{2.942 \text{ mW}}. \tag{36}$$

The average supply current was obtained from the current drawn by the pull-up side of the driver. The corresponding pull-down currents provide a consistency check for the current flowing through the differential output and load path.

## 1.10 Summary of Low-Swing Driver Results

The main results obtained for the optimized low-swing voltage-mode driver are summarized in Table 2.

Table 2: Summary of the low-swing voltage-mode driver results.

| Parameter | Result |
|---|---|
| Supply voltage $V_{DD}$ | 500 mV |
| Data rate | 8 Gb/s |
| Unit interval $T_{\text{UI}}$ | 125 ps |
| Pull-up multiplier $M_{PU}$ | 115 |
| Pull-down multiplier $M_{PD}$ | 52 |
| Pull-up $S_{11}$ at 4 GHz | $\approx -6.19$ dB |
| Pull-down $S_{11}$ at 4 GHz | $\approx -6.25$ dB |
| Differential maximum | $\approx +261.07$ mV |
| Differential minimum | $\approx -261.09$ mV |
| Differential peak-to-peak amplitude | $\approx 522.16$ mV$_{pp}$ |
| Average supply current | 5.884 mA |
| Average power | 2.942 mW |

## 1.11 Comment on the Low-Swing Driver Results

The S-parameter sizing simulations show that an optimum transistor size exists for both the pull-up and pull-down devices. Increasing the multiplier initially improves the impedance matching, while further increasing the device size introduces additional parasitic capacitance that degrades the high-frequency response. The optimum values obtained were $M_{PU} = 115$ and $M_{PD} = 52$.

The pull-up devices require a larger multiplier than the pull-down devices because the NMOS pull-up path experiences a reduction in effective $V_{GS}$ as the output voltage rises, reducing its available drive strength.

Using the optimum sizes in the complete driver resulted in a differential output of approximately $\pm 261$ mV, corresponding to a differential peak-to-peak amplitude of approximately 522 mV. The eye diagram confirms a clear and approximately symmetric differential signal at the target data rate.

The measured average supply current was 5.884 mA, giving an average power consumption of

$$\boxed{P_{\text{avg}} = 2.942 \text{ mW}}$$

at a 500 mV supply.

Overall, the low-swing driver achieves a substantial differential output swing while operating from a low supply voltage, making it attractive for high-speed serial-link applications where reducing voltage swing and power consumption is important.

# 2 Part 2: High-Swing Voltage Mode Driver

## 2.1 Objective

The second part of the lab investigates the high-swing voltage-mode driver. Unlike the low-swing driver of Part 1, the high-swing driver uses a higher supply voltage of

$$V_{DD} = 1 \text{ V}. \tag{37}$$

The objective is to optimize the pull-up transistor sizing using $S_{11}$ at the frequency of interest, then use the optimum sizing in the complete driver and evaluate its transient response, differential output swing, eye diagram, and average power consumption.

## 2.2 High-Swing Driver Architecture

The high-swing driver uses complementary MOS transistors. The pull-up devices are PMOS transistors, while the pull-down devices are NMOS transistors.

The two differential output branches are:

$$V_{DD} \to M_1 \to V_{OUTP} \to M_2 \to GND \tag{38}$$

and

$$V_{DD} \to M_0 \to V_{OUTN} \to M_3 \to GND. \tag{39}$$

The input control is differential, with $VINP$ and $VINN$ representing complementary PRBS signals. The PMOS devices provide the pull-up action, while the NMOS devices provide the pull-down action.

The differential output is defined as

$$V_{ODIFF} = V_{OUTP} - V_{OUTN}. \tag{40}$$

When $VINP$ and $VINN$ are complementary, the two output branches switch in opposite directions, producing a differential output signal.

Figure 10: Schematic of the high-swing voltage-mode driver.

## 2.3 Pull-Up Sizing Testbench

A separate sizing testbench was constructed to determine the optimum size of the pull-up PMOS transistor. The testbench includes the transistor under test, the 50 Ω port, the output capacitance, and the required bias sources.

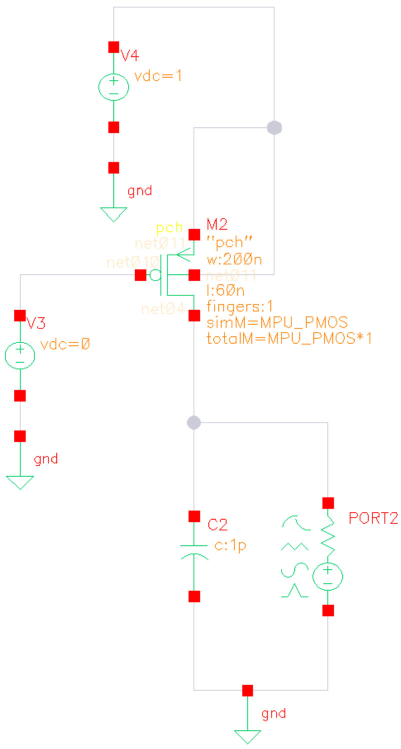


Figure 11: Sizing testbench used to determine the optimum pull-up PMOS transistor multiplier.

The purpose of the sizing procedure is to obtain the transistor multiplier that provides the best high-frequency matching at the frequency relevant to the 8-Gb/s data rate.

For an 8-Gb/s NRZ signal, the Nyquist frequency is

$$f_N = \frac{R_b}{2} = \frac{8 \text{ Gb/s}}{2} = 4 \text{ GHz}. \tag{41}$$

## 2.4 $S_{11}$ Versus Frequency

The first simulation was performed by sweeping the frequency in the S-parameter analysis. The purpose of this simulation was to determine the value of $S_{11}$ at the target frequency of 4 GHz before performing the transistor-size optimization.

The simulated $S_{11}$ response is shown in Figure 12.

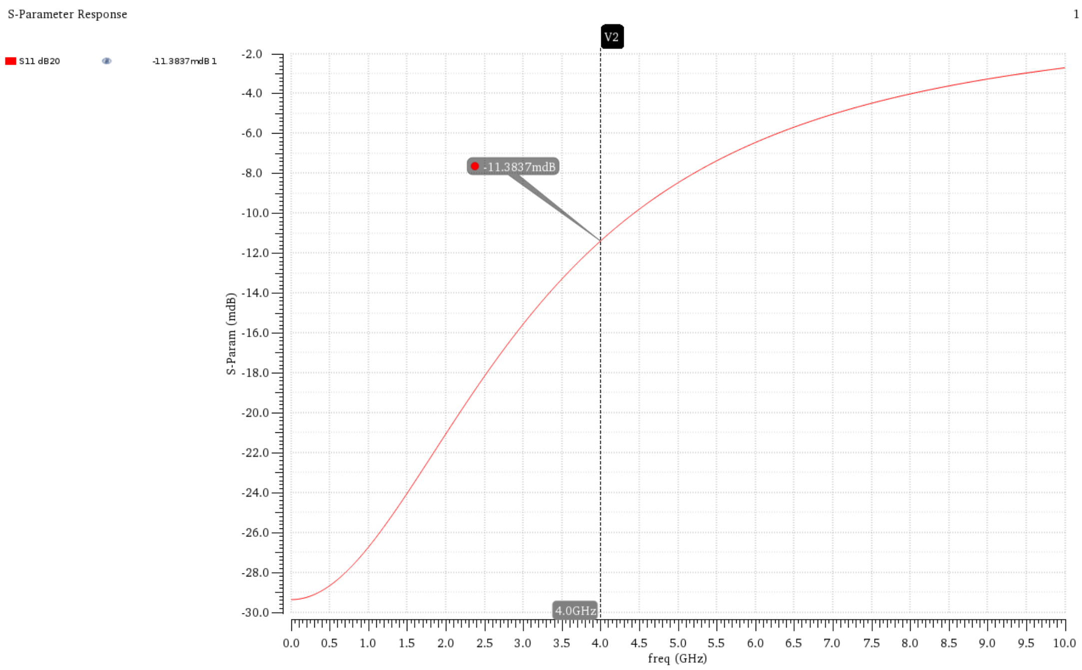


Figure 12: $S_{11}$ response versus frequency for the high-swing pull-up sizing testbench.

At

$$f = 4 \text{ GHz}, \tag{42}$$

the cursor gives

$$S_{11} \approx -11.3837 \text{ mdB}. \tag{43}$$

Since

$$1 \text{ dB} = 1000 \text{ mdB},$$

this corresponds to

$$S_{11} \approx -0.01138 \text{ dB}. \tag{44}$$

This result represents the initial matching condition before optimizing the pull-up transistor multiplier.

The $S_{11}$ curve is frequency dependent. As the frequency increases, the effect of the transistor parasitic capacitances and the output loading becomes more significant, causing the input impedance and consequently the reflection coefficient to vary with frequency.

## 2.5 $S_{11}$ Versus PMOS Multiplier

After identifying 4 GHz as the target frequency, the PMOS multiplier was swept in order to improve the matching at this frequency.

The multiplier was swept from

$$M_{PU} = 250 \rightarrow 350 \tag{45}$$

and the value of $S_{11}$ at 4 GHz was monitored for each multiplier value. The resulting $S_{11}$ versus multiplier curve is shown in Figure 13.

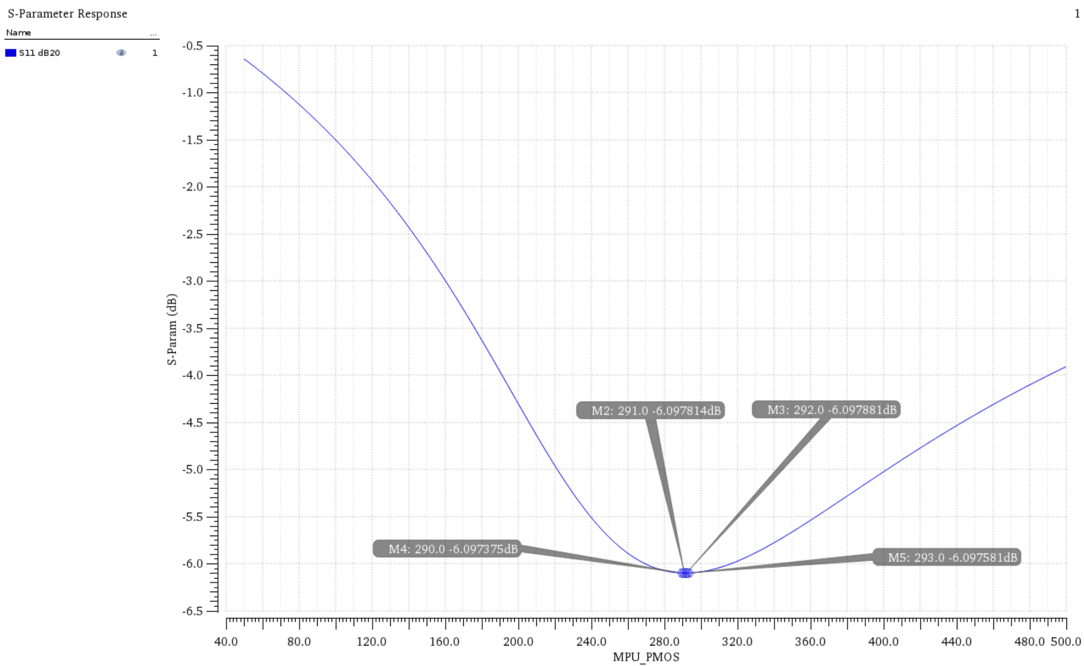


Figure 13: $S_{11}$(4 GHz) versus PMOS multiplier for the high-swing driver.

The curve exhibits a minimum around $M = 292$. The measured values near the minimum are approximately

$$M = 290 : -6.097375 \text{ dB}, \tag{46}$$
$$M = 291 : -6.097814 \text{ dB}, \tag{47}$$
$$M = 292 : -6.097881 \text{ dB}, \tag{48}$$
$$M = 293 : -6.097581 \text{ dB}. \tag{49}$$

Therefore, the optimum PMOS multiplier was selected as

$$\boxed{M_{PU} = 292} \tag{50}$$

with

$$\boxed{S_{11}(4 \text{ GHz}) \approx -6.0979 \text{ dB}}. \tag{51}$$

The minimum is relatively broad around the optimum, meaning that small changes around $M = 292$ have only a minor effect on the matching. Nevertheless, $M = 292$ gives the most negative measured $S_{11}$ and was therefore selected for the final high-swing driver.

## 2.6 Return to the Complete High-Swing Driver

After determining the optimum pull-up multiplier, the value

$$M_{PU} = 292 \tag{52}$$

was used in the complete high-swing voltage-mode driver.
The driver was then simulated using the complementary PRBS input signals at

$$R_b = 8 \text{ Gb/s} \tag{53}$$

with

$$T_{\text{UI}} = 125 \text{ ps.} \tag{54}$$

A transient simulation was performed for

$$T_{\text{sim}} = 50 \text{ ns.} \tag{55}$$

## 2.7 Differential Output and Peak-to-Peak Amplitude

The differential output was measured using

$$V_{ODIFF} = V_{OUTP} - V_{OUTN}. \tag{56}$$

The simulated differential output waveform is shown in Figure 14.

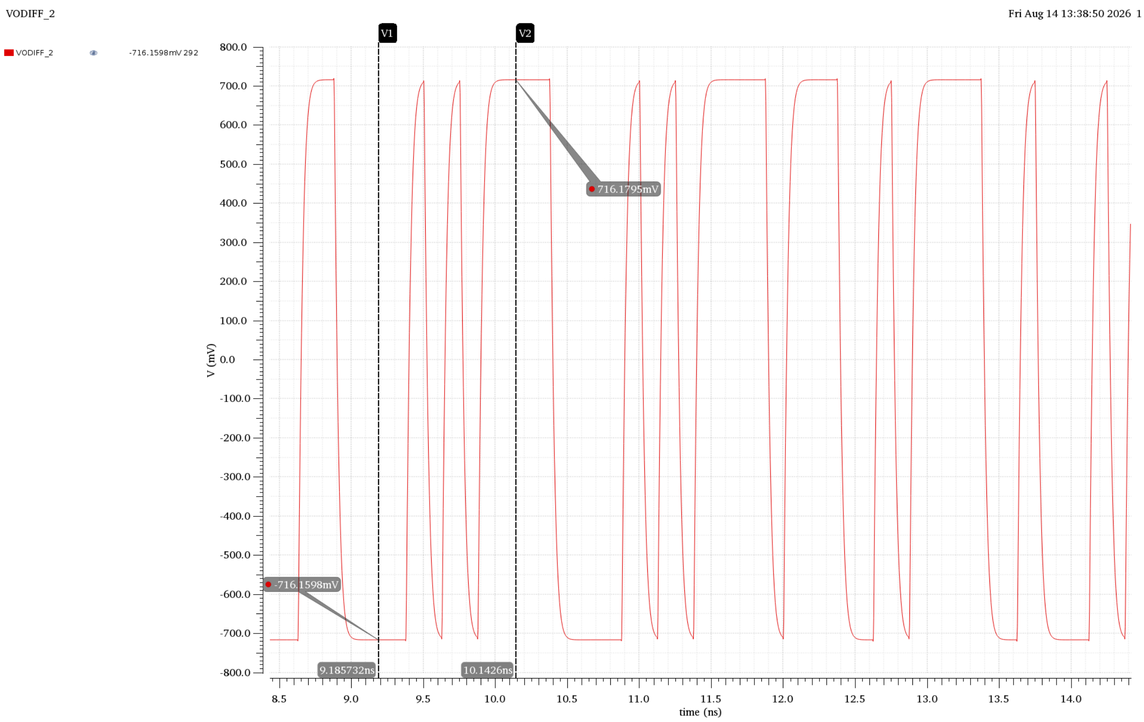


Figure 14: Transient response of the differential output of the optimized high-swing voltage-mode driver.

From the simulation, the maximum and minimum differential output levels are approximately

$$V_{ODIFF,\max} \approx +716.18 \text{ mV}, \tag{57}$$
$$V_{ODIFF,\min} \approx -716.16 \text{ mV}. \tag{58}$$

Therefore, the differential peak-to-peak amplitude is

$$V_{ODIFF,pp} = V_{ODIFF,\max} - V_{ODIFF,\min} \tag{59}$$
$$= 716.18 - (-716.16) \tag{60}$$
$$\approx 1432.34 \text{ mV}. \tag{61}$$

Hence,

$$\boxed{V_{ODIFF,pp} \approx 1.43 \text{ V}_{pp}}. \tag{62}$$

Although the supply voltage is 1 V, the differential peak-to-peak voltage can approach approximately $2V_{DD}$ because the two single-ended outputs switch in opposite directions. However, the actual differential swing is lower than the ideal $2V_{DD}$ value because of the finite transistor resistance and the output loading.

## 2.8 Eye Diagram

The eye diagram was generated from the differential output waveform using a unit interval of

$$T_{\text{UI}} = 125 \text{ ps}. \tag{63}$$

The simulated eye diagram is shown in Figure 15.

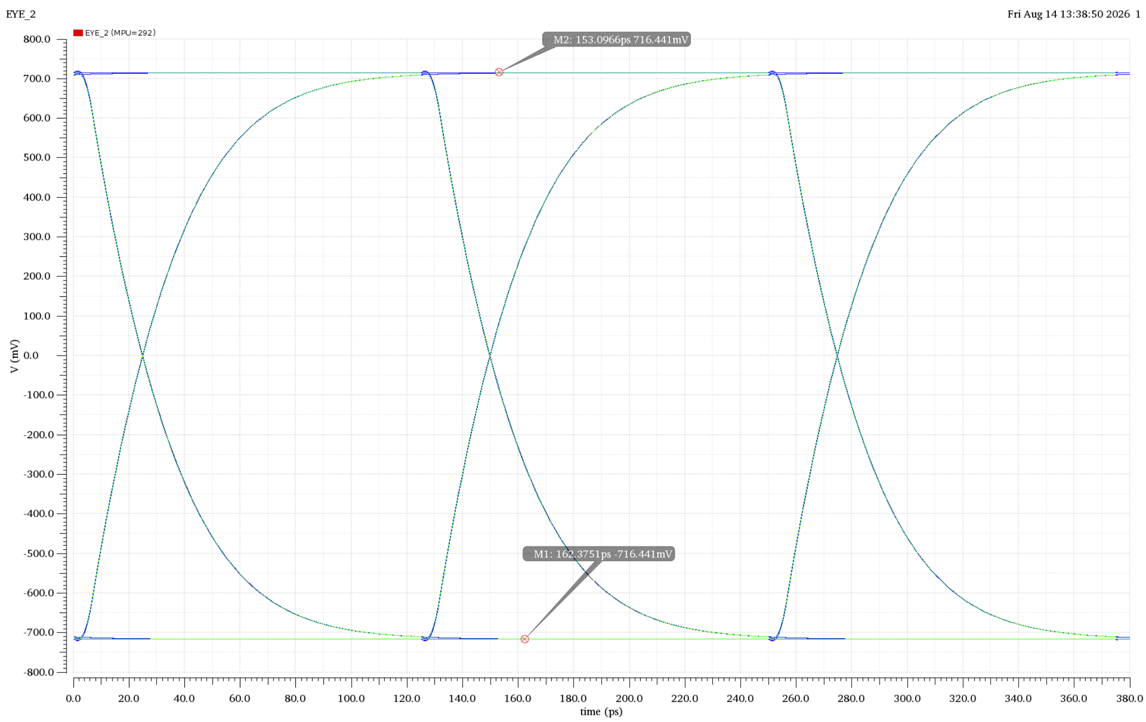


Figure 15: Eye diagram of the optimized high-swing voltage-mode driver at 8 Gb/s.

The upper and lower differential levels are approximately ±716 mV. Therefore, the observed vertical eye opening is approximately

$$H_{\text{eye}} \approx 716 - (-716) \approx \boxed{1.43\text{ V}}. \tag{64}$$

The eye is clearly open, demonstrating that the optimized high-swing driver produces a large differential voltage margin at the driver output. The rounded transition behavior is mainly due to the finite output resistance and the capacitive loading of the driver.

## 2.9 Average Power Consumption

The average power consumption was calculated according to

$$P_{\text{avg}} = V_{DD} I_{DD,\text{avg}}. \tag{65}$$

For the high-swing driver,

$$V_{DD} = 1\text{ V}. \tag{66}$$

The simulation result gives

$$\boxed{P_{\text{avg}} = 15.87\text{ mW}}. \tag{67}$$

The corresponding average supply current is therefore

$$I_{DD,\text{avg}} = \frac{P_{\text{avg}}}{V_{DD}} = \frac{15.87\text{ mW}}{1\text{ V}} = 15.87\text{ mA}. \tag{68}$$

The average-power measurement obtained from the simulation is shown in Figure 16.

| | | |
|---|---|---|
| LAB_03:High_swing_VM_Driver:1 | Pavg | 15.87m |
| LAB_03:High_swing_VM_Driver:1 | /M1/S | |
| LAB_03:High_swing_VM_Driver:1 | /M0/S | |

Figure 16: Average-power measurement for the optimized high-swing voltage-mode driver.

## 2.10 Summary of High-Swing Driver Results

The main simulation results for the high-swing driver are summarized in Table 3.

Table 3: Summary of the high-swing voltage-mode driver results.

| Parameter | Result |
|---|---|
| Supply voltage $V_{DD}$ | 1 V |
| Data rate | 8 Gb/s |
| Unit interval $T_{\mathrm{UI}}$ | 125 ps |
| Initial $S_{11}$ at 4 GHz | $-11.3837$ mdB |
| Initial $S_{11}$ at 4 GHz | $-0.01138$ dB |
| Optimum PMOS multiplier | 292 |
| Optimized $S_{11}$ at 4 GHz | $\approx -6.0979$ dB |
| Differential maximum | $\approx +716.18$ mV |
| Differential minimum | $\approx -716.16$ mV |
| Differential peak-to-peak amplitude | $\approx 1.43$ V$_{pp}$ |
| Average power | 15.87 mW |
| Average supply current | 15.87 mA |

## 2.11 Comment on the High-Swing Driver Results

The high-swing driver was first characterized using an S-parameter frequency sweep in order to identify the $S_{11}$ value at the target frequency of 4 GHz. After identifying the target frequency, the PMOS multiplier was swept to improve the matching at this frequency.

The multiplier sweep resulted in an optimum value of

$$\boxed{M_{PU} = 292}$$

with an $S_{11}$ value of approximately

$$\boxed{-6.10 \text{ dB}}.$$

The optimized transistor sizing was then applied to the complete driver and a 50-ns transient simulation was performed. The differential output reached approximately $\pm 716$ mV, resulting in a differential peak-to-peak amplitude of approximately

$$\boxed{1.43 \text{ V}_{pp}}.$$

The eye diagram shows a clear and large differential opening, while the average power consumption was

$$\boxed{15.87 \text{ mW}}.$$

Compared with the low-swing driver, the high-swing driver provides a larger differential output swing, but this comes at the expense of higher power consumption & area. This illustrates the fundamental trade-off between signal amplitude and power consumption in high-speed voltage-mode drivers.

## 2.12 Reasons for Preferring Low-Swing Operation

In some high-speed applications, a low-swing voltage-mode driver is preferred over a high-swing driver for the following reasons:

1. **Lower power consumption:**

   The dynamic switching power is approximately given by

   $$P_{\text{dyn}} \approx \alpha C V_{\text{swing}}^2 f. \tag{69}$$

   Therefore, reducing the voltage swing significantly reduces the dynamic power consumption because of the quadratic dependence on $V_{\text{swing}}$.

2. **Higher-speed operation:**

   A smaller voltage swing reduces the amount of charge that must be transferred to and from the output capacitance during each transition. This can reduce the transition time and facilitates high-speed operation at multi-Gb/s data rates.

3. **Potentially lower implementation area:**

   In some designs, the reduced swing can allow smaller driver devices and therefore lower transistor area. However, the actual area depends on the required output drive strength, impedance matching, and transistor sizing.